

Adaptive Signal Acquisition

A Living Technical Report on Sparse Reconstruction and Sequential Sensing

Project technical record

Version 3 — Stages 1 through 4B — 27 September 2026

Abstract

This report develops a reproducible simulation study of sparse-signal acquisition under a limited noisy-measurement budget. The central question is whether a sequential sensing policy, which chooses each new measurement after seeing earlier observations, improves reconstruction over fair non-adaptive baselines. Stages 1 and 2 establish the deterministic simulator, LASSO reconstruction, and fixed and random references. Stage 3 introduces a Gaussian information score augmented by a provisional-LASSO magnitude proxy. Stage 4A shows that the proxy is harmful: it rewards alignment with the current estimate rather than uncertainty. Stage 4B replaces it with an observation-dependent bootstrap covariance of plausible sparse reconstructions. In 100 paired trials the new method removes the large failure of the first adaptive policy and is statistically competitive with random and Gaussian-only sensing, but it does not yet provide a significant improvement. The report states the mathematical model, derives each score and update, defines every experimental quantity, records negative results, and identifies the next controlled experiments.

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1 Research question and scope

The primary question is:

When only a limited number of noisy linear measurements may be taken, can selecting each new measurement using the previous action–observation history reconstruct a sparse signal more accurately than selecting all measurements in advance?

The operational endpoint is normalized mean squared error (NMSE). A comparison is fair only when the methods share the hidden-signal distribution, candidate measurement dictionary, noise law, measurement budget, final reconstruction algorithm, and paired random draws. The project currently studies a synthetic, directly sparse vector. It is a controlled algorithmic test bed, not yet a claim about a particular physical sensor.

2 Notation and mathematical model

2.1 Notation

Symbol	Definition
N	Dimension of the unknown signal; $N = 64$ in the primary experiments.
k	Number of non-zero coordinates; $k = 4$ in the primary experiments.
$x \in \mathbb{R}^N$	Hidden signal. It is available to the simulator but never to a sensing policy.
$\text{supp}(x)$	Support of x , namely the indices of its non-zero entries.
$D \in \mathbb{R}^{M \times N}$	Candidate measurement dictionary; $M = 128$ normalized rows.
$a_t \in \mathbb{R}^N$	Column-form measurement vector selected at sequential step t ; $a_t^\top$ is a row of D .
B	Measurement budget, or number of selected rows.
$A_t \in \mathbb{R}^{t \times N}$	Sensing matrix formed by stacking the first t selected rows.
$y_t \in \mathbb{R}$	Scalar noisy observation returned for action a_t .
$y_{1:t} \in \mathbb{R}^t$	Vector containing the first t observations.
ϵ_t	Independent Gaussian measurement noise with variance σ^2 .
$\hat{x}_t$	Provisional sparse reconstruction after t observations.
μ_t, Σ_t	Mean and covariance of the Gaussian uncertainty approximation.
λ	LASSO sparsity-penalty coefficient; it is unrelated to the sensing weight γ .
γ	Stage 3/4A weight on the provisional-LASSO magnitude proxy.
R	Number of bootstrap reconstructions in Stage 4B; $R = 12$.
ρ	Stage 4B Gaussian shrinkage weight; $\rho = 0.25$.

2.2 Sparse hidden signal

A vector is k -sparse when exactly k coordinates are non-zero:

$$x \in \mathbb{R}^N, \quad \|x\|_0 = |\text{supp}(x)| = k \ll N. \quad (1)$$

The notation $\|x\|_0$ counts non-zero entries; it is not a norm in the strict algebraic sense. In each trial, the simulator samples k support indices uniformly without replacement and assigns independent amplitudes from $\mathcal{N}(0, 1)$. More generally a signal may be sparse in a known basis Ψ , so that $x = \Psi z$ and z is sparse. The current implementation uses $\Psi = I$, hence $z = x$.

2.3 Candidate actions and noisy measurements

Each candidate row contains N independent Rademacher entries (-1 or $+1$ with equal probability) and is normalized to unit Euclidean length. Thus

$$D = \begin{bmatrix} d_1^\top \\ \vdots \\ d_M^\top \end{bmatrix} \in \mathbb{R}^{M \times N}, \quad \|d_j\|_2 = 1, \quad M = 128. \quad (2)$$

The candidate count M is not the signal dimension N and is not the budget B . It is the number of allowed experiments from which a policy chooses. If indices $i_1, \dots, i_B$ are selected, the sensing matrix is

$$A_B = \begin{bmatrix} a_1^\top \\ a_2^\top \\ \vdots \\ a_B^\top \end{bmatrix} = \begin{bmatrix} d_{i_1}^\top \\ d_{i_2}^\top \\ \vdots \\ d_{i_B}^\top \end{bmatrix} \in \mathbb{R}^{B \times N}. \quad (3)$$

At step t , the environment returns one scalar:

$$y_t = a_t^\top x + \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma = 0.10. \quad (4)$$

Stacking the observations gives the standard linear inverse problem

$$y_{1:B} = A_B x + \epsilon. \quad (5)$$

A measurement is therefore a weighted sum of all coordinates, not necessarily a direct sample of one coordinate. The simulation represents what a physical linear measurement device would do after calibration supplies its response coefficients.

3 Stage 1: deterministic foundations

Stage 1 creates x , D , a fixed action subset, A , clean measurements Ax , a Gaussian noise vector, and observations y . A seeded NumPy generator makes the arrays exactly reproducible. The main shapes are

$$x : (64,), \quad D : (128, 64), \quad A : (B, 64), \quad y : (B,). \quad (6)$$

No reconstruction or adaptivity is used at this stage. Its purpose is to make every array in Equation (5) inspectable and to ensure that later differences come from algorithms rather than accidental changes in random data.

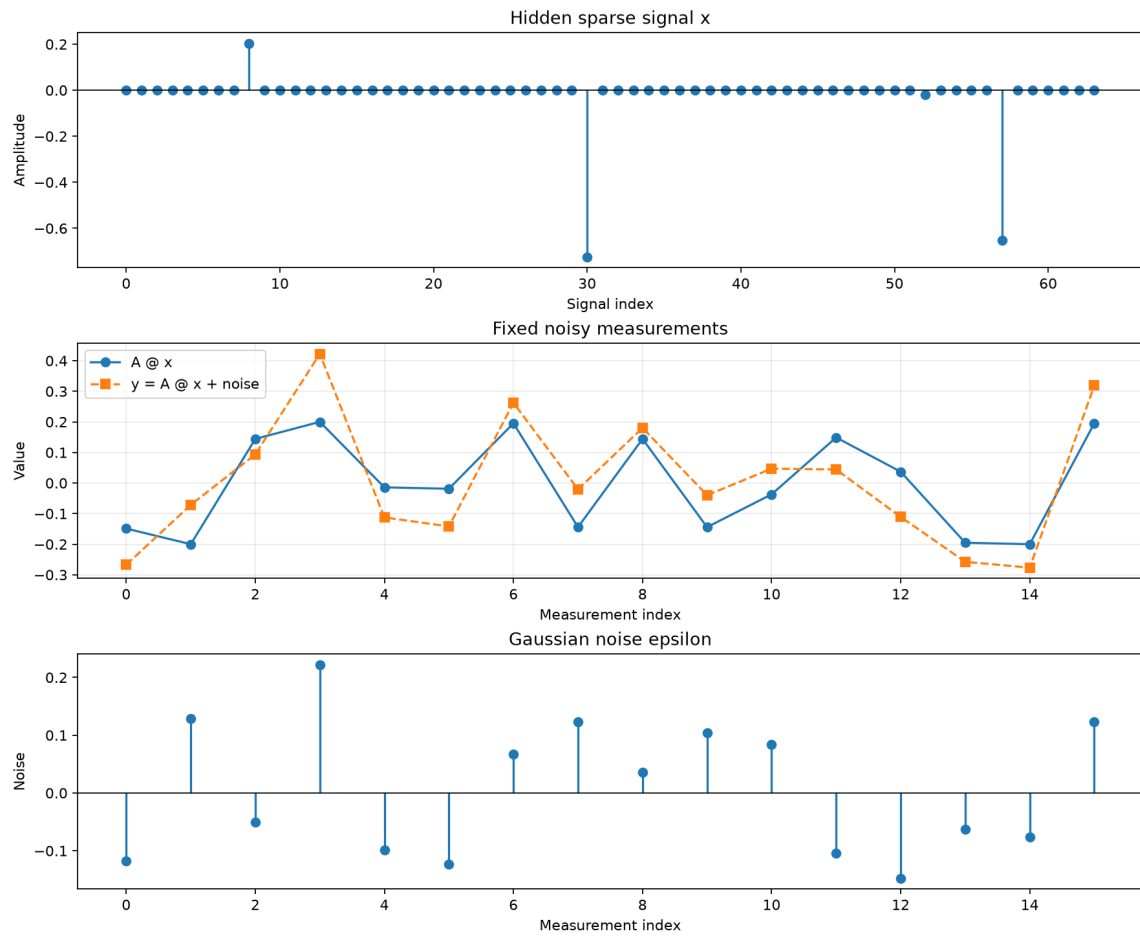


Figure 1: Stage 1 visual check of the sparse signal, clean measurements, noise, and observations.

4 Stage 2: sparse reconstruction and non-adaptive baselines

4.1 LASSO objective

After B measurements the project reconstructs the signal with LASSO:

$$\hat{x} = \arg \min_{z \in \mathbb{R}^N} \left\{ \frac{1}{2} \|A_B z - y_{1:B}\|_2^2 + \lambda \|z\|_1 \right\}. \quad (7)$$

Here z is a variable representing a candidate reconstruction; it is not the observed vector y . The first term is small when z explains the data. The second term, $\|z\|_1 = \sum_{j=1}^N |z_j|$, discourages many non-zero coefficients. The primary final solver uses $\lambda = 0.08$, at most 5000 iterations, and tolerance 10^{-7} .

4.2 ISTA derivation

Let $f(z) = \frac{1}{2} \|A_B z - y\|_2^2$. Its gradient is

$$\nabla f(z) = A_B^T (A_B z - y). \quad (8)$$

The gradient is Lipschitz with constant $L = \|A_B\|_2^2$, where $\|A_B\|_2$ is the largest singular value. Iterative shrinkage-thresholding (ISTA) performs

$$u^{(q)} = z^{(q)} - \frac{1}{L} \nabla f(z^{(q)}), \quad (9)$$

$$z^{(q+1)} = \mathcal{S}_{\lambda/L}(u^{(q)}), \quad (10)$$

where soft thresholding acts coordinatewise:

$$\mathcal{S}_\theta(u_j) = \text{sign}(u_j) \max\{|u_j| - \theta, 0\}. \quad (11)$$

The gradient step improves data agreement; soft thresholding is the proximal step for the ℓ_1 term and sets sufficiently weak coordinates to zero.

4.3 Fixed and random reference policies

The *fixed* policy uses the same precomputed row order in every trial. The *random* policy draws a row order before observing any y_t values. Both are non-adaptive. Within a trial all methods receive the same hidden x and the same noise prefix; only their selected rows differ. A budget $B = 16$ means 16 total measurements, not four groups of 16 and not 16 observations repeated across trials.

4.4 Metrics

The principal metric is

$$\text{NMSE}(x, \hat{x}) = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2 + 10^{-12}}. \quad (12)$$

The 10^{-12} term is only a numerical guard. Lower NMSE is better. Support recovery is reported using precision, recall, and their harmonic mean:

$$F_1 = \frac{2PR}{P+R}, \quad P = \frac{|\hat{S} \cap S|}{|\hat{S}|}, \quad R = \frac{|\hat{S} \cap S|}{|S|}, \quad (13)$$

where $S = \text{supp}(x)$ and $\hat{S} = \{j : |\hat{x}_j| > 10^{-3}\}$.

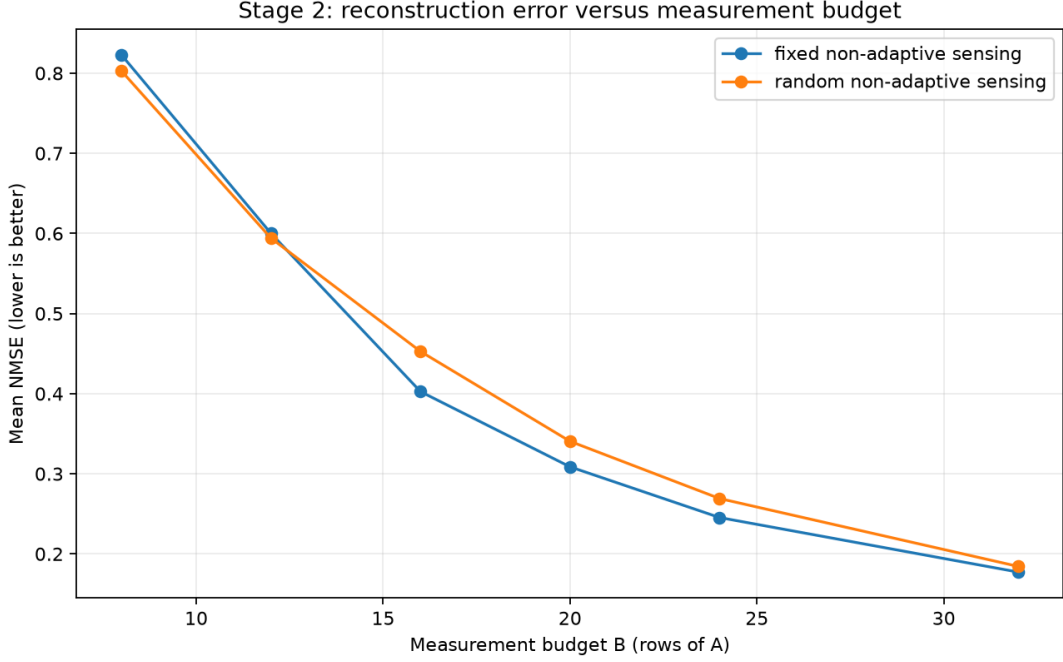


Figure 2: Stage 2 baseline behavior. Reconstruction error decreases as the budget grows.

5 Stage 3: first sequential information-guided policy

5.1 History and causal selection

At time t , the available history is

$$h_t = (A_t, y_{1:t}). \quad (14)$$

A valid policy selects $a_{t+1} = \pi(h_t)$ before the simulator evaluates $a_{t+1}^\top x + \epsilon_{t+1}$. The selector has no hidden-signal argument. This ordering prevents truth leakage.

5.2 Gaussian information score

Assume temporarily that the conditional belief is Gaussian:

$$x \mid h_t \sim \mathcal{N}(\mu_t, \Sigma_t). \quad (15)$$

For a candidate action a , the noiseless scalar $a^\top x$ has variance

$$\text{Var}(a^\top x \mid h_t) = \mathbb{E}[(a^\top (x - \mu_t))^2 \mid h_t] = a^\top \Sigma_t a. \quad (16)$$

Because independent noise adds variance, $y \mid h_t, a$ has variance $a^\top \Sigma_t a + \sigma^2$. The entropy of a scalar Gaussian with variance v is $\frac{1}{2} \log(2\pi e v)$. Hence the mutual information is

$$I(x; y \mid h_t, a) = H(y \mid h_t, a) - H(y \mid x, h_t, a) \quad (17)$$

$$= \frac{1}{2} \log \left(1 + \frac{a^\top \Sigma_t a}{\sigma^2} \right). \quad (18)$$

The implementation omits the positive factor $\frac{1}{2}$ because it cannot change the ranking. It selects the unused row maximizing

$$s_G(a) = \log \left(1 + \frac{a^\top \Sigma_t a}{\sigma^2} \right). \quad (19)$$

5.3 Gaussian conditioning update

After observing (a_t, y_t) , the exact linear-Gaussian update is

$$K_t = \frac{\Sigma_t a_t}{a_t^\top \Sigma_t a_t + \sigma^2}, \quad (20)$$

$$\mu_{t+1} = \mu_t + K_t(y_t - a_t^\top \mu_t), \quad (21)$$

$$\Sigma_{t+1} = \Sigma_t - K_t a_t^\top \Sigma_t. \quad (22)$$

K_t is the gain. The covariance update depends on the selected direction but not on the numerical observation. Therefore Gaussian-only sensing is sequential but, with fixed dictionary and tie breaking, its row sequence is not observation-value adaptive.

5.4 The adaptive_v1 magnitude proxy

To make Stage 3 react to y , a provisional LASSO estimate $\hat{x}_t$ was inserted through

$$\tilde{\Sigma}_t = \Sigma_t + \gamma \hat{x}_t \hat{x}_t^\top. \quad (23)$$

For a candidate a ,

$$a^\top \tilde{\Sigma}_t a = \underbrace{a^\top \Sigma_t a}_{g_t(a)} + \underbrace{\gamma (a^\top \hat{x}_t)^2}_{\ell_t(a)}. \quad (24)$$

Thus γ controls only the sensing score; it does not change the LASSO penalty λ . The selected action maximizes

$$s_{v1}(a) = \log \left(1 + \frac{g_t(a) + \ell_t(a)}{\sigma^2} \right). \quad (25)$$

6 Stage 3 result and Stage 4A diagnosis

The Stage 3 policy was worse than the non-adaptive references at every tested budget. Table 2 reports mean NMSE over 100 trials.

Table 2: Stage 3 mean NMSE; lower is better.

B	Fixed	Random	adaptive_v1
8	0.8324	0.8238	0.9006
12	0.5616	0.6010	0.7795
16	0.4045	0.4185	0.6499
20	0.3024	0.3396	0.5078
24	0.2400	0.2792	0.4028
32	0.1815	0.2004	0.2724

Stage 4A varied only $\gamma \in \{0, 0.1, 0.5, 1\}$ and recorded the separate components in Equation (24), row correlation, singular values, condition number, and provisional reconstruction quality. The complete 100-trial result is shown in Table 3.

At $B = 32$, Gaussian-only ($\gamma = 0$) had paired difference -0.0220 from random with 95% bootstrap interval $[-0.0424, -0.0011]$. The original $\gamma = 1$ policy had difference $+0.0649$ with interval $[+0.0277, +0.1007]$. Increasing γ raised the mean maximum row correlation from 0.17 to 0.22 and the mean condition number from 1.79 to 2.52.

Table 3: Stage 4A mean NMSE for the magnitude-proxy ablation.

B	Random	$\gamma = 0$	$\gamma = 0.1$	$\gamma = 0.5$	$\gamma = 1$
8	0.8263	0.7685	0.8197	0.8643	0.8441
12	0.5876	0.5649	0.6188	0.7497	0.7033
16	0.3750	0.3788	0.4489	0.6126	0.5589
20	0.2843	0.2607	0.3310	0.4868	0.4528
24	0.2251	0.2070	0.2473	0.3988	0.3669
32	0.1663	0.1443	0.1633	0.2681	0.2312

The diagnosis follows directly from Equation (24): $\hat{x}_t \hat{x}_t^\top$ represents estimated magnitude, not uncertainty. It rewards rows aligned with what the provisional estimate already believes and can repeatedly measure similar directions. A valid replacement should ask where plausible reconstructions *disagree*.

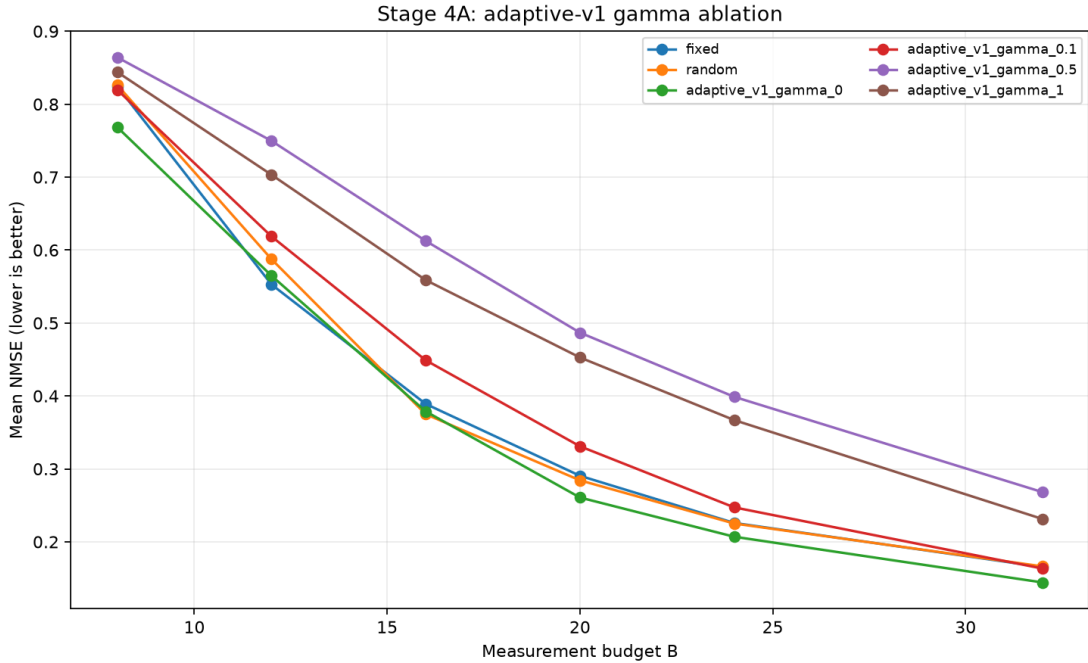


Figure 3: Stage 4A: larger magnitude-proxy weights degrade reconstruction.

7 Stage 4B: adaptive_v2 bootstrap uncertainty

7.1 Design requirements

The replacement was implemented as a separate policy; adaptive_v1 remains unchanged for comparison. The new policy must (i) depend on observed values, (ii) represent uncertainty rather than magnitude, (iii) retain geometric diversity, (iv) never receive the hidden signal, and (v) be exactly reproducible under a fixed policy seed.

7.2 Gaussian-diverse burn-in

Bootstrap uncertainty cannot be estimated before observations exist. For the first $b_0 = 8$ actions, `adaptive.v2` therefore uses Equation (19). This burn-in collects diverse rows and produces an initial history $(A_{b_0}, y_{1:b_0})$. It is not random: it greedily covers directions that remain uncertain under Σ_t .

7.3 Parametric bootstrap ensemble

At a post-burn-in step t , generate $R = 12$ perturbed observation vectors

$$y_{1:t}^{(r)} = y_{1:t} + \eta^{(r)}, \quad \eta^{(r)} \sim \mathcal{N}(0, \tau^2 \sigma^2 I_t), \quad r = 1, \dots, R, \quad (26)$$

with perturbation multiplier $\tau = 1$. For each perturbation, solve

$$\hat{x}_t^{(r)} = \arg \min_z \left\{ \frac{1}{2} \|A_t z - y_{1:t}^{(r)}\|_2^2 + \lambda \|z\|_1 \right\}. \quad (27)$$

The implementation evaluates all R ISTA problems in one vectorized matrix update. The policy-internal solver uses at most 500 iterations and tolerance 10^{-5} ; the final evaluation still uses the unchanged 5000-iteration LASSO.

Define the ensemble mean and empirical covariance by

$$\bar{x}_t = \frac{1}{R} \sum_{r=1}^R \hat{x}_t^{(r)}, \quad (28)$$

$$\hat{\Sigma}_{B,t} = \frac{1}{R-1} \sum_{r=1}^R (\hat{x}_t^{(r)} - \bar{x}_t)(\hat{x}_t^{(r)} - \bar{x}_t)^\top. \quad (29)$$

For any candidate a ,

$$a^\top \hat{\Sigma}_{B,t} a = \frac{1}{R-1} \sum_{r=1}^R \left[a^\top (\hat{x}_t^{(r)} - \bar{x}_t) \right]^2. \quad (30)$$

Equation (30) is the sample variance of the measurement predicted by the ensemble. A large value means plausible sparse explanations of the existing data predict different outcomes for action a . That is the intended notion of informativeness.

7.4 Trace matching and shrinkage

The Gaussian and bootstrap covariances can have very different total scales. Directly adding them would make their relative influence depend on solver scale rather than an explicit design parameter. The implementation trace-matches the bootstrap covariance:

$$\hat{\Sigma}_{B,t}^{\text{scaled}} = \frac{\text{tr}(\Sigma_t)}{\max\{\text{tr}(\hat{\Sigma}_{B,t}), \varepsilon_{\text{num}}\}} \hat{\Sigma}_{B,t}, \quad (31)$$

where ε_{num} is machine-scale protection against division by zero. If the empirical covariance is numerically zero, the policy falls back to Gaussian-only sensing. Otherwise it forms

$$\Sigma_{v2,t} = \rho \Sigma_t + (1 - \rho) \hat{\Sigma}_{B,t}^{\text{scaled}}, \quad \rho = 0.25. \quad (32)$$

The Gaussian term supplies a full-rank diversity reference. This matters because an R -member sample covariance has rank at most $R - 1 = 11$, much smaller than $N = 64$.

7.5 Action score and selection

Every unused candidate is scored by

$$s_{v2}(a) = \log \left(1 + \frac{a^\top \Sigma_{v2,t} a}{\sigma^2} \right), \quad a_{t+1} = \arg \max_{a \in \mathcal{D} \setminus \mathcal{U}_t} s_{v2}(a), \quad (33)$$

where $\mathcal{D}$ is the candidate set and $\mathcal{U}_t$ contains already used actions. The selected raw variance is recorded as separate Gaussian and bootstrap contributions:

$$\rho a^\top \Sigma_t a + (1 - \rho) a^\top \hat{\Sigma}_{B,t}^{\text{scaled}} a. \quad (34)$$

Unlike the Gaussian covariance, $\hat{\Sigma}_{B,t}$ is a function of $y_{1:t}$. LASSO is nonlinear because soft thresholding can change active sets, so changing the observations can change both the ensemble support patterns and covariance. A unit test explicitly verifies that two observation histories produce different bootstrap covariances while using the same perturbation draws.

7.6 End-to-end algorithm

For one trial, `adaptive.v2` performs the following causal loop:

1. Initialize $\mu_0 = 0$ and $\Sigma_0 = (4/64)I = 0.0625I$.
2. For $t < 8$, score unused rows with Gaussian variance only.
3. For $t \geq 8$, construct the bootstrap ensemble from $(A_t, y_{1:t})$, compute Equations (29)–(32), and score unused rows.
4. Select the maximum-score row. Only now does the simulator evaluate $y_{t+1} = a_{t+1}^\top x + \epsilon_{t+1}$.
5. Append the row and observation, update the Gaussian belief, and repeat.
6. At each requested budget, reconstruct with the common final LASSO and compute metrics.

8 Stage 4B experiment and result

8.1 Frozen protocol

The study used 100 paired trials, $N = 64$, $k = 4$, $M = 128$, $\sigma = 0.10$, budgets $B \in \{8, 12, 16, 20, 24, 32\}$, final $\lambda = 0.08$, and seed 20260927. Each method was run once to $B = 32$ per trial; smaller budgets reuse exact prefixes. Five policies were compared: fixed, random, Gaussian-only, frozen `adaptive.v1` with $\gamma = 1$, and `adaptive.v2`. Confidence intervals are deterministic percentile-bootstrap intervals from 2000 resamples of the paired trial-level NMSE differences.

Table 4: Stage 4B mean NMSE over 100 paired trials.

B	Fixed	Random	Gaussian-only	adaptive_v1	adaptive_v2
8	0.7956	0.7983	0.8210	0.8815	0.8210
12	0.5959	0.5243	0.5547	0.6967	0.5627
16	0.3894	0.3775	0.3437	0.5759	0.3807
20	0.2904	0.2807	0.2707	0.4643	0.2748
24	0.2289	0.2281	0.2231	0.4005	0.2228
32	0.1541	0.1657	0.1480	0.2788	0.1586

Table 5: Paired adaptive_v2 NMSE difference from random; negative favors adaptive_v2.

B	Mean difference	95% lower	95% upper
8	+0.0227	-0.0469	+0.0940
12	+0.0384	-0.0312	+0.1111
16	+0.0033	-0.0534	+0.0651
20	-0.0059	-0.0513	+0.0388
24	-0.0053	-0.0469	+0.0410
32	-0.0071	-0.0383	+0.0332

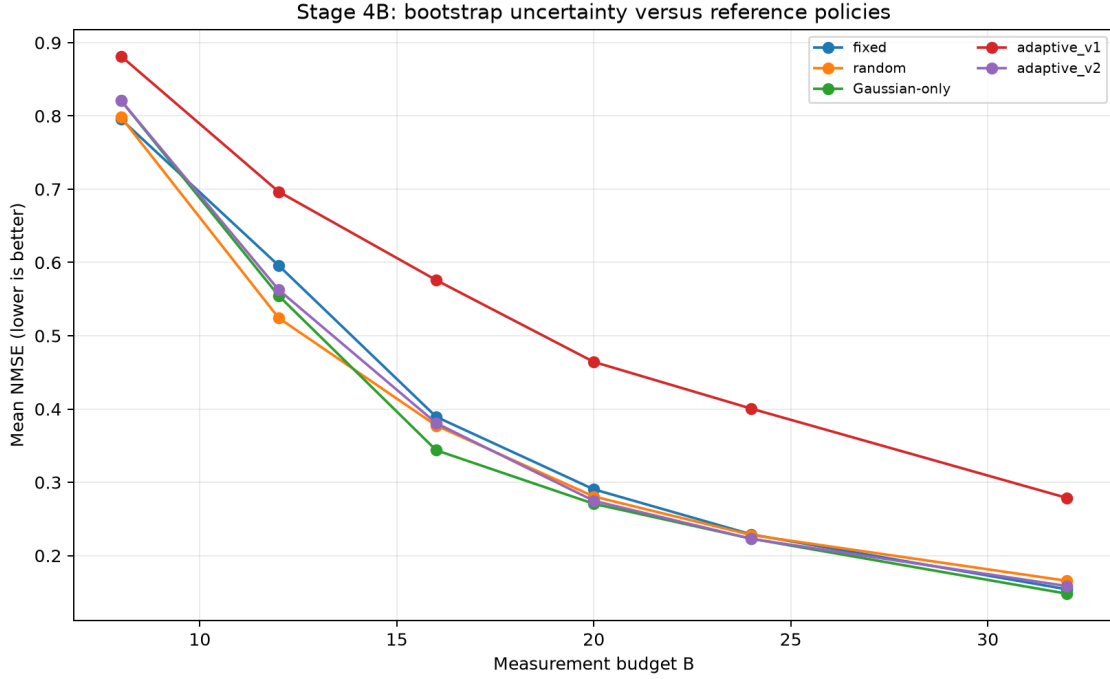


Figure 4: Stage 4B comparison. adaptive_v2 removes the large adaptive_v1 degradation but does not separate significantly from random or Gaussian-only sensing.

8.2 Interpretation

The result supports three precise statements.

1. **The replacement fixed the identified failure mode.** At $B = 32$, adaptive_v1 is worse than random by $+0.1132$ with interval $[+0.0745, +0.1534]$, whereas adaptive_v2 differs by -0.0071 with interval $[-0.0383, +0.0332]$.
2. **Observation-dependent uncertainty did not yet produce a demonstrated gain.** Every adaptive_v2 interval in Table 5 crosses zero. It is therefore accurate to call the method competitive, not superior.
3. **The tested dense random dictionary leaves limited room for adaptivity.** Fixed, random, Gaussian-only, and adaptive_v2 become close at moderate and large budgets. Dense normalized rows already provide broad coverage, so selecting among them may matter less than it would for structured dictionaries or clustered supports.

Across the 2400 post-burn-in decisions, the bootstrap term supplied a mean 0.8705 fraction of the selected raw variance, the empirical covariance had mean rank 10.98 (consistent with the theoretical maximum 11), mean maximum row correlation was 0.25, mean condition number was 2.83, and mean post-observation provisional NMSE was 0.3389. These diagnostics show that the bootstrap term is active; the null result is not caused by silently falling back to Gaussian-only sensing.

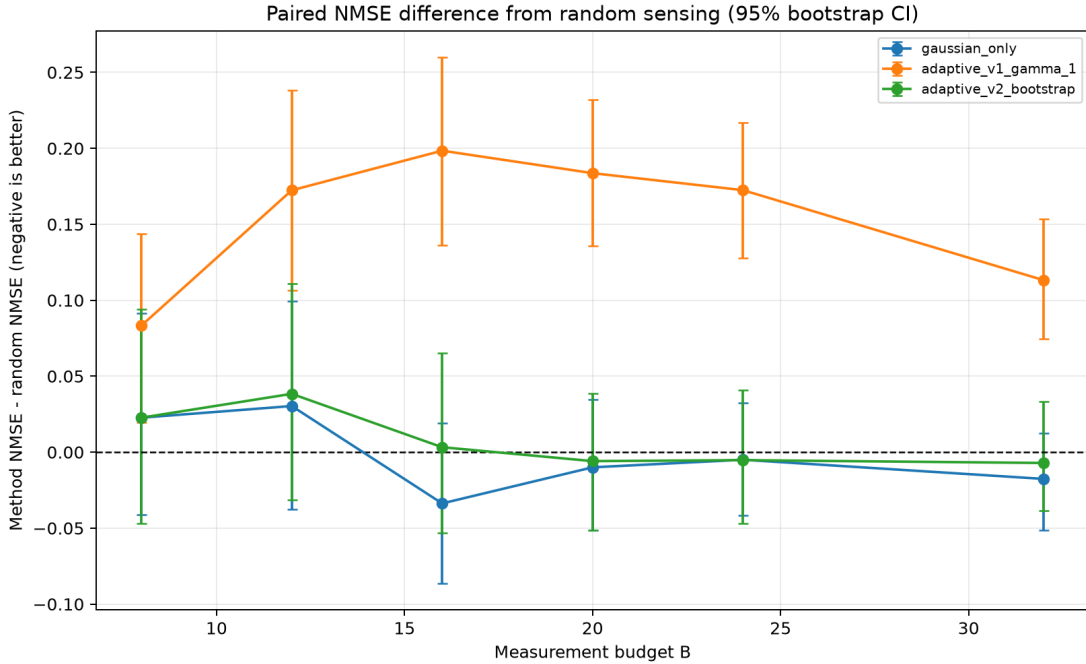


Figure 5: Paired NMSE differences from random sensing with 95% bootstrap intervals.

9 What should happen next

Stage 4B is complete as an implementation and base-setting evaluation. It should not yet be followed by an LLM orchestrator, because the scientific action space is not sufficiently understood. The next work should be a controlled Stage 4C sensitivity study:

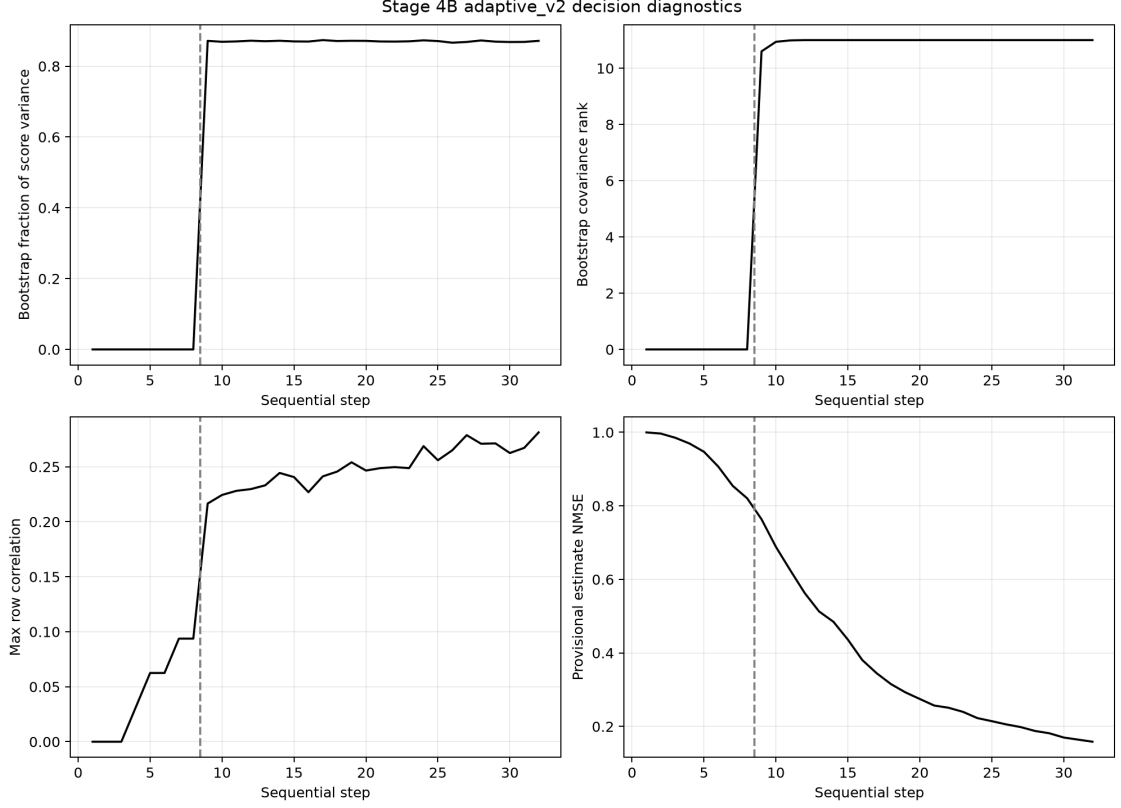


Figure 6: Decision diagnostics. The dashed vertical line separates the eight-step burn-in from bootstrap-guided sensing.

1. vary $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ to measure the diversity–bootstrap trade-off;
2. vary burn-in length and ensemble size R ;
3. compare observation perturbation with residual and wild bootstrap variants;
4. test lower SNR, different sparsity levels, clustered supports, and structured dictionaries in which adaptive selection has a clearer opportunity to help;
5. pre-register a primary setting and confirm selected conclusions with 500 trials.

Only after this policy family and experimental registry are stable should an LLM manage experiment selection, summarize evidence, or propose bounded follow-up configurations. The LLM should not replace the numerical policy, the reconstructor, or the deterministic result registry.

10 Implementation map and reproducibility

Path	Responsibility
<code>signals.py</code>	Sparse-signal generation.
<code>measurements.py</code>	Candidate dictionary and $y = Ax + \epsilon$.
<code>reconstruction.py</code>	LASSO objective and ISTA solver.
<code>baselines.py</code>	Fixed and random action orders.
<code>sequential.py</code>	Gaussian belief and frozen adaptive_v1.
<code>stage4.py</code>	Stage 4A gamma ablation and diagnostic records.
<code>adaptive_v2.py</code>	Bootstrap ensemble, covariance, shrinkage, and sequential loop.
<code>stage4b.py</code>	Paired Stage 4B experiment and bootstrap confidence intervals.
<code>run_stage4b_study.py</code>	CSV, metadata, and figure generation.
<code>tests/test_stage4b.py</code>	Determinism, causality, observation dependence, and paired-study tests.

Run the verification and primary study from the repository root:

```
.venv\Scripts\python.exe -m pytest -q
.venv\Scripts\python.exe scripts\run_stage4b_study.py
.venv\Scripts\python.exe scripts\build_living_report.py
```

The Stage 4B output directory is `artifacts/stage4b_adaptive_v2/`. It contains trial-level records, paired summaries, decision traces, exact metadata, and figures. Random seeds are separated for dictionary, fixed order, signal, noise, random baseline, bootstrap policy, and confidence-interval resampling. Runtime naturally varies; selected actions and quality metrics reproduce.

11 Limitations

The current evidence is limited to directly sparse Gaussian-amplitude signals, a dense Rademacher candidate dictionary, one noise level, and LASSO reconstruction. The bootstrap ensemble perturbs observations conditionally on one measured history and does not represent all forms of model uncertainty. Trace matching is an explicit heuristic, not a Bayesian identity. The final numerical conclusions apply to the frozen configurations reported here; they do not establish that adaptive sensing is generally useless or generally superior.

References

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